

AI Hotel Workforce Management System

3-Phase Product Requirements Document (PRD)

This PRD defines a phased implementation strategy for a template-driven, multi-tenant SaaS platform for hotels, resorts, restaurants, and banquet operations. The system combines HRMS, payroll, compliance, finance, GST, inventory, subscription billing, mobile applications, and advanced AI.

1. Core Architectural Principles

- Multi-tenant SaaS architecture
- Template-driven configuration through Settings and Template Configuration Center
- Role-Based Access Control (RBAC)
- API-first architecture with REST APIs and webhooks
- Configurable approval workflows and automation rules
- Modular architecture with phased delivery

2. Phase 1 – Build All Major Modules

Objective: Launch a production-ready web platform by implementing the first working version of all major business modules except Stores & Inventory Management, Mobile Applications, and advanced AI.

- Platform Foundation & Administration
- Settings with Template Configuration Center
- HR Activities Module
- Finance & Accounting Module
- GST & Tax Engine
- Dashboards, Reports & Analytics
- Automation & Notifications
- SaaS Billing & Subscription Module
- Integrations Module (core integrations only: ZKTeco, Razorpay/Cashfree, Google Maps, WhatsApp, Email/SMS)
- API & Developer Platform
- Security, Privacy & Governance

3. Phase 2 – Enhance All Modules + Stores Management + Mobile Applications

Objective: Refine all modules built in Phase 1, add enterprise enhancements, and implement the complete Stores & Inventory Management module and mobile applications.

- Advanced improvements and refinements across all Phase 1 modules

- Complete Stores & Inventory Management Module
- Warehouse, Godown, and Store Management
- Stock Scanning & Barcode Management
- Batch, Lot, Expiry, and Wastage Tracking
- Purchase Orders and Goods Received Notes (GRN)
- POS Billing and Employee Purchases with Payroll Deduction
- Employee Mobile App (Android & iOS)
- Employer/Admin Mobile App
- Offline Attendance
- Mobile Approvals
- Digital ID Cards

4. Phase 3 – Advanced AI, CCTV/Webcam Analytics, and Enterprise Integrations

Objective: Implement advanced AI capabilities, computer vision, predictive analytics, and optional enterprise hospitality integrations.

- Face Recognition and Liveness Detection
- CCTV/Webcam Attendance
- Idle Time Detection
- Break Duration Analysis
- Unauthorized Absence Detection
- Productivity Alerts and Performance Scorecards
- Absenteeism Prediction
- Attrition Prediction
- Payroll Anomaly Detection
- Occupancy Staffing Forecast
- Sentiment Analysis
- HR Chatbot
- AI Insights Dashboard
- Oracle OPERA PMS Integration (optional)
- Cloudbeds PMS Integration (optional)
- Banking API Integrations

5. Strategic Benefits of This Phased Approach

- **Phase 1** — delivers a commercially usable web platform quickly.
 - **Phase 2** — adds inventory and mobile after core workflows are stabilized.
 - **Phase 3** — introduces AI after sufficient operational data has been collected.
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